

## Week 4

### Practical Exercises

You are working on a software development project to build a software for management of students' records:

1. Create a class called "StudentRecord", the information assigned for a student are their name, their student ID, their year of enrolment, and the duration of the programme they are registered on.
2. For a given student, the code should be able to get the student's name given their ID and vice versa (Make these getter functions)
3. For a given student, the code should have the ability to give the status of the student (Enrolled or Graduated). A student is enrolled if:  
$$2024 - \text{the year of enrolment} \leq \text{the duration of the programme}$$
4. Create another class called "ModuleRecord" to keep a record of all students registered on a module, a student can be added to the module by a line of code like:

`ELEC362 += MohammadHasan;`

The class should also allow for deletion using the "-" operator. And it should allow the code to list all students registered on a given module.

It is part of the exercise to think of what data members and function members of the classes are. There are multiple ways to do this task. Consider what the advantages and the disadvantages of each.

Hint: You may need to use the `push_back()` function we developed in an earlier lab exercise session, you can find its implementation in the code W4.cpp with this lab session document.

Hint: You will need to define a zero parameter constructor of the class `StudentRecord` for the `push_back()` function to work.